

CONSTRUCTION OF A REALISTIC RED-BLOOD-CELL COMPUTATIONAL MODEL AT THE MOLECULAR LEVEL

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ABSTRACT

Biological membranes have been studied using computational models that span different spatial and temporal scales. Continuum models capture large-scale membrane behavior using effective physical properties [1]. At a finer level of resolution, atomistic molecular dynamics represents individual atoms and their interactions within the membrane. However, this level of detail limits simulations to smaller systems and shorter timescales [2]. Between these two levels of description, coarse-grained models reduce molecular detail while retaining important membrane properties.

In this study, we used the coarse-grained membrane model developed by Fu et al. [3]. Each membrane particle represents a collection of lipids with a characteristic length scale of approximately 5 nm. The membrane particles are not connected by permanent bonds. Instead, they interact through a pair potential that depends on both the distance and relative orientation. To represent a red blood cell, we combined this membrane model with a coarse-grained cytoskeleton and explicit internal and external water. The cytoskeleton includes junction complexes, spectrin tetramers, and ankyrin particles. Spectrin tetramers are represented as chains of consecutively bonded beads. Harmonic bonds hold the cytoskeleton network together and connect it to the membrane. Water particles interact through Lennard-Jones potentials. Our goal is to build a red blood cell with a realistic diameter of about 7 μm and to reproduce its characteristic biconcave shape. The simulation starts from a spherical configuration containing internal water particles. We then progressively reduce the effective size of these water particles, which decreases the enclosed volume and drives the shape transition. In preliminary simulations with smaller systems, the biconcave shape was easily obtained. However, the transition becomes more difficult when the model is scaled up to a realistic cell size. This is because a larger system containing more particles can explore more configurations before reaching the biconcave state. To address this, we implemented a global restraint on the total membrane area of the model. This restraint applies a quadratic penalty whenever the area deviates from the reference value. As a result, the enclosed volume can decrease while the membrane area stays nearly constant. We implemented the corresponding forces in LAMMPS [4]. In addition, we

observed that pores appeared in some realistic-size models even before the volume reduction began, allowing internal water particles to escape. This separate stability problem is currently being studied by adjusting the membrane-particle density and the other model parameters.

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